



Multiple Choice:

1. When the temperature increases, the intrinsic stand-off ratio _____.
A. increases
B. decreases
C. essentially remains the same
D. Either A or B
2. What device does not have a gate terminal?
A. Triac
B. SCR
C. FET
D. Diac
3. An SCR is made of silicon and not germanium because silicon _____.
A. is inexpensive
B. is mechanically strong
C. has low leakage current
D. is tetravalent
4. Between the peak point and valley point of UJT emitter characteristic we have _____ region.
A. saturation
B. negative-resistance
C. cut-off
D. None of the above
5. Which of the following conditions is necessary for triggering system for thyristors?
A. It should be synchronized with the main supply
B. It must use separate power supply
C. It should provide a train of pulses
D. None of these
6. When a thyristor gets turned ON, the gate drive
A. Should not be removed as it will turn-off the SCR
B. May or may not be removed
C. Should be removed
D. Should be removed in order to avoid increased losses and higher junction temperature
7. The V-I characteristics for a triac in the first and third quadrants are essentially identical to those of _____ in the first quadrant.
A. SCR
B. UJT
C. transistor
D. SCS
8. Which of the following PNP devices has two gates?
A. Triac
B. SUS
C. SCS
D. Diac
9. After peak point, the UJT operates in the _____ region
A. cut off
B. saturation
C. negative-resistance
D. Either A or B
10. A SCR is rated at 75A peak, 20A average. The greatest possible delay in the trigger angle if the dc is at rated value is
A. 47.5°
B. 45.7°
C. 48.5°
D. 43.7°
11. When the temperature increases, the inter base-resistance (R_{BB}) of a UJT _____.
A. remains the same
B. decreases
C. increases
D. Either A or B
12. Which of the following types of heating process is used for surface heating of steel?
A. Dielectric heating
B. Infra-red heating
C. Induction heating
D. Resistance heating
13. An SCR has _____ pn junctions.
A. two
B. four
C. three
D. five
14. The UJT may be used as _____.
A. an amplifier
B. a sawtooth generator
C. a rectifier
D. a multivibrator
15. In normal operation of an SCR, gate is _____ with respect to cathode.
A. positive
B. negative
C. at zero potential
D. Either A or B
16. We can control a.c. power in a load by connecting _____.
A. two SCRs in series
B. two SCRs in parallel
C. two SCRs in parallel opposition
D. Either A or B
17. A transistor PWM chopper supplies power to a parallel combination of inductive load and freewheel diode. The battery supply is 24 V, the load has a resistance of 1 and inductance of 10 mH, and the chopper frequency is 100 kHz. Determine the average load voltage for a turn-on time of 8 ms.
A. 19.2 V
B. 21.6 V
C. 12.9 V
D. 14.5 V
18. The intrinsic stand-off ratio (η) of a UJT is given by _____.
A. $\frac{R_{B1}}{R_{B1} + R_{B2}}$
B. $\frac{R_{B1} R_{B2}}{R_{B1}}$
C. $\frac{R_{B1} R_{B2}}{R_{B2}}$
D. $R_{B1} R_{B2}$
19. For thyristors, pulse triggering is preferred to dc triggering because
A. gate dissipation is low
B. pulse system is simpler
C. triggering system is require for a very short duration
D. All of these
20. The normal way to turn on a diac is by _____.
A. break over voltage



- B. gate current
C. gate voltage
D. Either B or C
21. Cycloconverter converts _____
- A. ac voltage to dc voltage
B. dc voltage to dc voltage
C. ac voltage to ac voltage at same frequency
D. ac voltage at supply frequency to ac voltage at load frequency
22. A UJT is sometimes called _____ diode.
- A. double-based
B. low resistance
C. high resistance
D. single base
23. An SCR has _____ semi-conductor layers.
- A. four
B. two
C. three
D. five
24. If gate current is increased, then anode-cathode voltage at which SCR closes _____.
- A. remains the same
B. is increased
C. is decreased
D. None of the above
25. The SCR is turned – off when the anode current falls below
- A. forward current rating
B. breakover voltage
C. holding current
D. latching current
26. A diac has _____.
- A. one pn junction
B. two pn junction
C. three pn junctions
D. four pn junctions
27. The normal way to turn on an SCR is by _____.
- A. breakover voltage
B. appropriate gate current
C. appropriate anode current
D. None of the above
28. In dc choppers, the waveform for
- A. input voltage is continuous and output voltage is discontinuous
B. input voltage is discontinuous and output voltage is continuous
C. input voltage as well as output voltage both are continuous
D. input voltage as well as output voltage both are discontinuous
29. A triac is a _____ switch.
- A. unidirectional
B. bidirectional
C. mechanical
D. electronic
30. A step-up chopper is to provide a 48 V across a 12 Ω load from a 12 V DC supply. The chopper inductance is 10 H and the chopper frequency is 100 kHz. Determine the duty cycle required.
- A. 0.75
B. 0.65
C. 0.95
D. 0.85
31. Power electronics essentially deal with the control of a.c. power at _____.
- A. frequencies above 20 kHz.
B. frequencies above 1000 kHz
C. frequencies less than 10 Hz
D. 50 Hz frequency
32. Which of the following PNP devices has a terminal for synchronizing purpose?
- A. SCS
B. Diac
C. Triac
D. SUS
33. In a GTO, anode current begins to fall when gate current
- A. is negative peak at time $t = 0$
B. is negative peak at time $t =$ storage time t_s
C. just begins to become negative at $t = 0$
D. None of these
34. In dc chopper, per unit ripple is maximum when duty cycle α is
- A. 0.2
B. 0.7
C. 0.5
D. 0.8
35. An SCR behaves as a _____ switch.
- A. bidirectional
B. unidirectional
C. mechanical
D. electronic
36. Peak inverse rating of a triac
- A. is the same as that of a thyristor
B. is greater than that of a thyristor
C. is inferior and very significant due to the nature of its application
D. is not very significant due to the nature of its application
37. In a SCR circuit, the angle of conduction can be changed by changing the _____.
- A. anode voltage
B. anode current
C. forward current rating
D. gate current
38. A diac has _____ terminals.
- A. two
B. four
C. three
D. five
39. If gate current is increased, the anode-cathode voltage at which SCR closes is _____.
- A. increased
B. decreased
C. maximum
D. least
40. In a thyristor dc chopper, which type of the commutation results in best performance?
- A. Voltage commutation
B. Current commutation
C. Load commutation
D. Supply commutation



41. A triac can pass a portion of _____ half cycle through the load.

- A. only positive
- B. only negative
- C. Both positive and negative
- D. Either A or B

42. The normal way to close a SCR is by approximate

- A. gate current
- B. cathode current
- C. anode current
- D. forward current

43. A cycloconverters can be considered to be composed of two converters

- A. Connected back to back
- B. Series connected
- C. Parallel connected
- D. Series-parallel connected

44. An SCR is sometimes called a _____.

- A. triac
- B. thyristor
- C. diac
- D. unijunction transistor

45. A triac is equivalent two SCRs _____.

- A. in parallel
- B. in series
- C. in inverse-parallel
- D. Either A or B

46. An amplifying gate thyristor has

- A. the advantage of high current at low level gate drive
- B. a poor di/dt rating even at high gate current
- C. its di/dt improving only at high gate current
- D. very slow spreading velocity

47. What device exhibits negative resistance region?

- A. Diac
- B. SCR
- C. Triac
- D. UJT

48. A 60 V chopper drives a resistive load of $12\ \Omega$. The chopper frequency is set to 1 kHz. Calculate the values of rms load voltage and average load current for a turn-on time of 0.2 ms.

- C. 2.24 A
- D. 2.46 A
- C. 2.52 A
- D. 2.12 A

49. The turn-off gain of the GTO is of the order of

- A. 1-2
- B. 10-20
- C. 3-5
- D. >100

50. In the normal operation of an SCR, anode is _____ with respect to cathode.

- A. negative
- B. at zero potential
- C. positive
- D. Either A or C

END